

Spatiotemporal variation in drivers of parasitoid metacommunity structure in continuous forest landscapes

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Abstract. Although landscape spatial structure is known to influence spatial patterns of biodiversity, its effect on insect communities at higher trophic levels such as parasitoids remains poorly understood. This is particularly true in continuously distributed forests in which it can be difficult to identify clear boundaries among habitat patches. Using the metacommunity framework, we evaluate the relative importance of forest landscape structure, non-environmental spatial structure, and host outbreak status to spatial and within-season temporal variation in parasitoid communities. We used variation partitioning and metacommunity structure analyses to identify (1) the drivers of the metacommunity structure of parasitoids associated with the spruce budworm (*Choristoneura fumiferana*), and (2) how their relative influence varies through a season. We used a multi-scale perspective to summarize landscape heterogeneity in regions of increasing size around the community sampling locations. Spruce budworm larvae and pupae were sampled during three periods during the summer 2014 in 18 locations within continuous forest landscapes in Quebec, Canada. Thirty-two parasitoid wasp and fly species were recorded, 16 of which were found at more than one location. We found that the mechanisms shaping metacommunity structure changed over the course of a single season and that community structure varied among sites. At early and late periods in the season, we found that non-environmental structure, forest structure, and likely inter-specific competition were the main mechanisms influencing spatial variation in community structure. These results suggest a competition–dispersal trade-off. In contrast, at the middle period of the season, environmental filtering by forest structure and stochastic events were found to influence community structure. This period corresponds to the transition between early and late parasitoid communities. Our findings on the role of environmental filtering and forest structure support the idea that forest manipulations have the potential to influence parasitoid populations and hence spruce budworm outbreak dynamics as hypothesized by the “enemies hypothesis.” Moreover, our study highlights the value of considering a multi-scale approach and temporal variability of species interactions when characterizing the multiple processes shaping spatial metacommunity structure, particularly in continuous environments.

Key words: boreal forest; continuous landscape; elements of metacommunity structure; multi-scale framework; spruce budworm, *Choristoneura fumiferana*; variation partitioning.

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INTRODUCTION

Spatial variation in forest composition and configuration (i.e., forest structure) affects biodiversity,

community dynamics, and ecosystem functioning through its effects on habitat availability, quality, and connectivity (Schmiegelow and Mönkkönen 2002, Fahrig 2003, Foley et al. 2005, Harper et al.

2005). Improving our understanding of how forest spatial structure affects ecological communities and the biodiversity that comprises them is a significant and important task in spatial ecology and conservation biology (Hooper et al. 2005, McCann et al. 2005, Srivastava and Vellend 2005, Tscharn-tke et al. 2012).

In the boreal forest, the relationship between forest structure and speciose communities of insect parasitoids remains poorly understood (Cronin and Reeve 2005, Fraser et al. 2007), despite the abundant biodiversity they represent and their important influence on the population dynamics of irruptive forest insect pests (Royama 1984, Eveleigh et al. 2007). Parasitoid population and community dynamics are impacted by the success of species searching behavior and attack rates, which are known to be affected by (1) the distribution and density of hosts (Morrison and Strong 1980); (2) forest spatial structure including that of the host's habitat (Elzinga et al. 2007); (3) the spatial scale at which the parasitoid perceives forest structure (Roland and Taylor 1997); and (4) the presence of competitors (Xu et al. 2016). However, the relative importance of each of these factors to generating spatial patterns in biodiversity and in affecting outbreak dynamics is unclear (Cronin and Reeve 2014). Improved understanding of these relationships is essential to better understand the complex spatial dynamics of host–parasitoid systems (Cronin and Reeve 2014), to provide new approaches to forest management, and to mitigate outbreak consequences. Such understanding is critical within the context of ongoing human activities that affect landscape fragmentation, forest composition, and climate, all of which are likely to affect spruce budworm attack rates and outbreak cycles in the future (Su et al. 1996, Cappuccino et al. 1998, Campbell et al. 2008, Pureswaran et al. 2015).

Forest insect outbreaks represent a major disturbance to northern forests (Volney and Fleming 2007) that have significant effects on ecosystem functioning and ecosystem service provisioning (Maguire et al. 2015, MacLean 2016). The “enemies hypothesis” suggests that forest habitat diversity can increase top-down trophic control of outbreaking populations through an increase in parasitoid diversity, abundance, and efficiency (Russell 1989, Jactel and Brockerhoff 2007). In turn, increased parasitism has the potential to attenuate the economic consequences of outbreaking pests such as the

eastern spruce budworm (*Choristoneura fumiferana* (Clem.); Lepidoptera: Tortricidae; Cappuccino et al. 1998). Although correlations have been found between stand-level hardwood content and parasitoid species richness (Eveleigh et al. 2007) and parasitism rates (Cappuccino et al. 1998, Quayle et al. 2003), we do not yet know precisely how forest structure affects communities of parasitoid natural enemies at different spatial scales.

The spruce budworm is a native, irruptive defoliator of balsam fir (*Abies balsamea*) and spruce (*Picea* spp.; Royama 1984) and is one of the most destructive native forest pests in North America, with recurring outbreaks ca. every 30 yr (Volney and Fleming 2007). The spruce budworm food web is complex and includes a diverse community of parasitoids, which varies with host density during outbreak periods (Eveleigh et al. 2007). The parasitoid community also varies in composition within a single season due to differences in parasitoid species phenology (Eveleigh and Johns 2014, Seehausen et al. 2016). Improved understanding of the drivers of spruce budworm-associated parasitoid abundance and community structure has the potential to contribute to ecosystemic forest management approaches to reduce the negative consequences of pest outbreaks, as has been achieved in agroecosystems (e.g., Krueess 2003, Tscharn-tke et al. 2005).

Metacommunity theory

Investigation into the drivers of spatial variation in forest communities requires awareness of the different processes involved and the multiple scales at which they operate (e.g., Ricklefs 1987, Borthagaray et al. 2015). However, community theory has traditionally focused on single, local scales and assumed that local communities are independent (Logue et al. 2011). Metacommunity theory, on the other hand, provides a spatially explicit and multi-scaled perspective on community dynamics in which local communities are connected through dispersal (Leibold et al. 2004). The metacommunity framework integrates information on regional species pools and the local biological processes that drive community assembly to improve our understanding of how biodiversity is distributed in space and time.

Metacommunity analyses employ multiple modeling approaches to characterize relationships between spatial structure and biodiversity to

identify the mechanisms involved in shaping community structure (e.g., Logue et al. 2011). Metacommunities are thought to be structured by three mechanisms (Leibold et al. 2004; Table 1): (1) species interactions, (2) dispersal, and (3) environmental filtering. Based on these mechanisms, four conceptual paradigms have been proposed to describe what shapes a metacommunity (Leibold et al. 2004; Table 1): (1) species sorting, (2)

mass effects, (3) patch dynamics, and (4) neutral dynamics. These paradigms differ in terms of the relative importance they give to each of the three metacommunity-shaping mechanisms. These paradigms do not cover all possibilities and are not mutually exclusive, as empirical studies often identify community structures that can be described by more than one paradigm or none of them (Logue et al. 2011, Brown et al. 2016).

Table 1. Definitions of metacommunity mechanisms and paradigms, and elements of metacommunity structure and associated theoretical patterns of species' distribution.

Term	Definition
Metacommunity mechanisms	
Environmental filtering	"Environmental conditions govern species' occurrence." (Biswas and Wagner 2012)
Species interactions	"Competition among species for resources or space governs species' occurrence. Other interactions, such as facilitation, are not commonly considered in metacommunity studies." (Biswas and Wagner 2012)
Dispersal	"[...] any movement of individuals or propagules with potential consequences for gene flow across space. Such definition thus includes both natal dispersal and breeding dispersal." (Ronce 2007)
Metacommunity paradigms (source: Logue et al. 2011)	
Species sorting	"[...] acts on the assumption that habitat patches differ with regard to environmental conditions. Where dispersal is not limited (i.e., species arrive at all habitat patches), differences in the ability to cope with these environmental conditions enable species to coexist regionally by means of niche diversification and differences in resource exploitation."
Mass effects	"[...] assumes that environmentally heterogeneous habitat patches are tightly interconnected via frequent dispersal in such a way that reproduction in a source habitat allows for persistence within a sink habitat."
Patch dynamics	"[...] paradigm assumes environmentally homogeneous patches to be inhabited by species that exhibit a trade-off between dispersal and local dominance, such as a colonisation – competition trade-off in which successful competitors are poor colonisers and vice versa."
Neutral	"[...] implies that species do not differ in their fitness and niche. Community assembly depends entirely on demographic stochasticity, and immigration and speciation is assumed to counteract local extinction processes."
Elements of metacommunity structure (source: Leibold and Mikkelsen 2002)	
Coherence of species ranges	"A completely coherent species range is an array of sites, all of which contain the species. In other words, it is a sequence of the species' "presences" that is not interrupted by any of the species' "absences". The more interruptions, the less coherent the range."
Species turnover	"[...] reflects the tendency for species to replace each other from site to site."
Boundary clumping	"[...] is the degree to which the boundaries of different species' ranges are clustered together."
Theoretical patterns of species' distribution among sites (source: Leibold and Mikkelsen 2002)	
Checkerboards	"Species pairs have mutually exclusive distributions (i.e., "forbidden combinations") but such pairs occur independently of other pairs."
Nested subsets	"Species list from different sites form a formal set of nested subsets."
Clementsian	"Gradients result in discrete communities that replace each other as a group."
Gleasonian	Gradients result in species turnover but the arrangement of species ranges along the gradient is random."
Evenly spaced gradients	"Gradients result in no discrete communities but species ranges are arranged more evenly than expected by random chance."
Random	"There are no gradients or other patterns in species distributions among sites."

Metacommunity framework in forest ecosystems

Parasitoid communities in forest ecosystems present three important challenges to metacommunity analysis. First, spatial variation in forest landscapes is continuous (Fahrig et al. 2011); hence, it is difficult to clearly define “patches” and to identify individual local communities. Such continuous variation differs from the traditional patch-matrix paradigm (*sensu* Forman 1995) in which habitat patches are thought to be distributed within an inhospitable landscape matrix and local communities delimited by those patches. Explicit consideration of continuous variation in forest structure has been shown to improve inferences regarding metacommunity-shaping processes (Biswas and Wagner 2012). However, it is not always clear at which spatial scale one should model the relationships between community diversity and spatial structure. One approach to resolve this second challenge involves taking a multi-scale perspective that allows one to ask not only which landscape attributes are influencing metacommunity structures, but also at what scale(s) they are operating (Meynard et al. 2013). Finally, as parasitoid systems are dynamic within and among years (Stireman and Singer 2002, Eveleigh et al. 2007), it is not clear how to include such temporal variation in an analysis of community structure and how this variation might confound metacommunity analyses if it is not taken into account. Although these three challenges have previously been addressed independently, to our knowledge they have not yet been considered all together in an integrated way using empirical data.

In this paper, we combine recent extensions of the existing metacommunity framework (e.g., Chisholm et al. 2010, Alexander et al. 2011, Fernandes et al. 2013, Bellmore et al. 2015) to investigate how spatial heterogeneity structures the spruce budworm-associated parasitoid metacommunity in a continuous forest landscape at three different points in time within a single season. Specifically, we ask the following two questions: (1) How and at which spatial scales do spatial autocorrelation, forest structure, and spruce budworm outbreak status shape parasitoid metacommunity structure? and (2) How do their relative influences vary through a season? To answer these questions, we used two complementary approaches that differ in their ability to identify

metacommunity structuring mechanisms: variance partitioning (VP) and the elements of metacommunity structure (EMS) approach. Variation partitioning can be used to distinguish community patterns that arise due to dispersal from those due to environmental filtering (Legendre and Legendre 2012). The EMS approach can be used to distinguish among environmental filtering, competition, and random assembly (Leibold and Mikkelsen 2002). Using both approaches, we are able to identify the extent to which metacommunity structure is created by each of the four hypothesized paradigms (Meynard et al. 2013) and infer the underlying role of each of the three mechanisms discussed above.

We expect that environmental filtering through landscape structure and host availability should influence parasitoid metacommunity structure. We also expect that dispersal would shape metacommunities, as parasitoids are thought to follow spruce budworm outbreak and aggregate in high-density areas (the so-called “bird-feeder effect”; Eveleigh et al. 2007). Finally, we expect that competition among parasitoid species will not play an important role in shaping metacommunity structure because of the non-limiting high density of hosts (i.e., spruce budworm larvae and pupae) in the study area.

MATERIALS AND METHODS

Study region and data collection

In summer 2014, 4.3 million ha of public forest in the province of Quebec, Canada, was being affected by spruce budworm defoliation (Quebec Government 2015a). The present study was carried out within and around the largest outbreak area, located in the Côte-Nord region of the province in the boreal and boreal-mixedwood forest (Fig. 1). Spruce budworm larvae and pupae were sampled from 18 sites, chosen on the basis of defoliation status, distance from other sampling sites (~40 km), and ease of access from logging roads.

It is known that the spruce budworm parasitoid community changes through time and that different species attack different developmental stages of the host (Eveleigh and Johns 2014). Because of these temporal changes, species involved in consecutive parasitoid communities may be affected by different environmental features. To characterize these differences, spruce

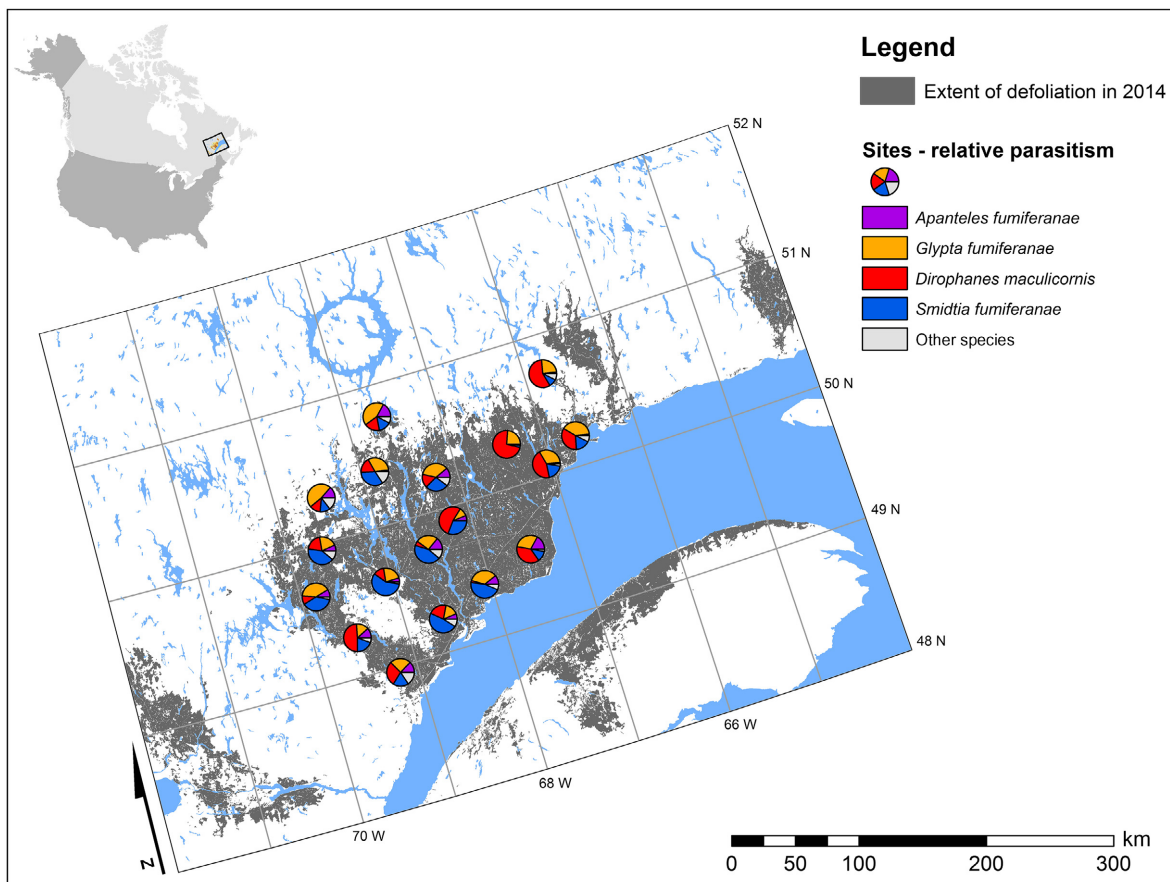


Fig. 1. Map of the study area in the Côte-Nord region in Quebec, Canada, where a massive spruce budworm outbreak takes place. Are represented the outbreak extent in 2014 (gray area) and our 18 sample sites (pie charts). Pie charts detail the proportion of relative parasitism over all spruce budworm stages by the four main parasitoid species and other species.

budworm individuals were collected at three different time periods between early June and mid-July, which correspond to three different stages of spruce budworm development: early larvae (EL; instars L3–L4), late larvae (LL; instars L5–L6), and pupae (P). Timing of field sampling was guided by the BioSIM 10 phenological model (Régnière et al. 2013), which uses accumulated degree days to predict current spruce budworm developmental stage. For each site and developmental stage, we collected ca. 500 larval and pupal specimens that were reared individually on a synthetic diet (McMorran 1965, Hervet et al. 2016). Emergent primary parasitoids were identified to species following McAlpine et al. (1981), O'Hara (2005), Bennett (2008), and Fernández-Triana and Huber (2010). Spruce budworm

larvae were identified on the basis of morphology (MacKay 1962, Harvey and Stehr 1967), host on which they were feeding (Stehr 1967), and development stage at the time of collection relative to predicted phenology (BIOSIM; Régnière et al. 2013). Thus, all brownish larvae having paired lateral white spots on each segment, found on either balsam fir, white spruce, or black spruce during our targeting sampling periods, were identified as *Choristoneura fumiferana*.

Per species relative parasitism rates were calculated for each site and developmental stage following Blais (1960). We first calculated the apparent parasitism for each species at each site and developmental stage, by dividing the number of emerged parasitoids of each species by the total number of spruce budworm host larvae and

pupae reared. We only included larvae and pupae that survived until either moth or parasitoid emergence; larvae and pupae that died for unknown reasons were not included as their parasitism status was unknown (Peralta et al. 2014). Per species relative parasitism rates for each developmental stage were then calculated by reducing apparent parasitism in proportion to the relative parasitism of previous stages (if present; Blais 1960). This approach controls for the larvae already parasitized at previous stages. We also calculated per species relative parasitism for each site over all stages.

The three resulting matrices (species $\times$ sites) of relative parasitism rates (one for each sample period) were used as the response variable in multivariate (i.e., RDA [Redundancy Analysis] for variation partitioning) and EMS analyses. Species that were reported at only one site were not retained for further analysis.

Three different types of predictor variables were used to model parasitism rates: (1) (multi-scale) forest structure; (2) spruce budworm outbreak status; (3) and spatial autocorrelation (non-environmental spatial structure). We describe each predictor type

in detail below. Using this approach, we sought to identify which environmental features shape meta-community structure and at which spatial scales.

Local and landscape forest attributes

Forest structure and spruce budworm cumulative defoliation were measured at multiple spatial scales, in six concentric buffers of increasing radius around each sample site (Fig. 2). Buffer radii examined were 100, 500, 1000, 2500, 5000, and 10,000 m. Geoprocessing and buffer extraction were undertaken using QUANTUMGIS 2.14 (QGIS Development Team 2016).

Forest structure.—Remotely sensed Landsat 8 satellite imagery (USGS) was used to characterize forest landscape structure in our study area using tiles from summer 2013 and 2014. Supervised classification of raw imagery was undertaken using the software PCI Geomatica 2015 and resulted in 23 initial forest land-cover types. Map resolution was improved from 30 m to 15 m using the panchromatic band using the PANSHARP function in PCI Geomatica 2015. The resulting land-cover map was then simplified to remove small patches (i.e., single 30×30 m pixels; 900 m^2) by

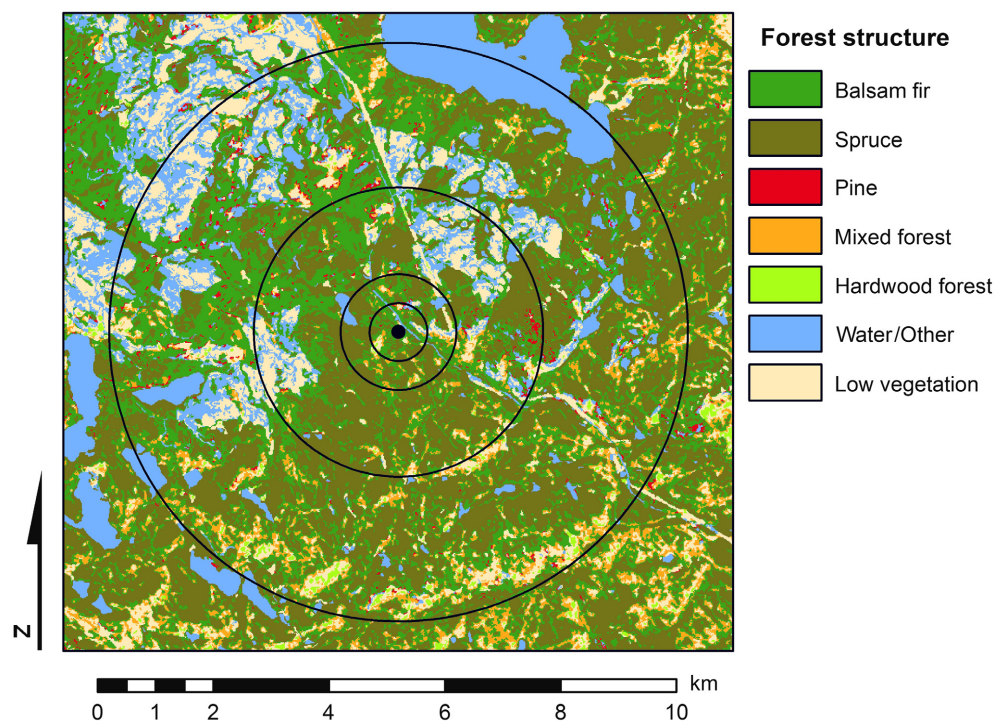


Fig. 2. Example of buffer areas used to characterize forest structure surrounding a sampling location.

merging them with the dominant surrounding patch. The final land-cover map was then reduced to six spruce budworm relevant land-cover types: balsam fir, spruce, pine, hardwood, mixed forest, and low vegetation (herb/grass; Table 2). This final map was used in all subsequent analyses (Pontbriand-Paré 2016).

Forest composition was summarized within each buffer as the proportion of each of the six cover types. We also calculated Shannon’s diversity index based on these cover types in each buffer (Table 2). Configuration was summarized within each buffer using three landscape pattern metrics calculated using FRAGSTATS 4.2 (McGarigal et al. 2012): edge density, contagion, and perimeter–area ratio (Table 2).

Spruce budworm outbreak status.—Local outbreak status was estimated using larval density during the first sampling period (EL stage). At each study site, 10 branches of 45 cm in length were collected from 10 different trees evenly distributed throughout the entire site. On each branch, we counted the number of larvae and annual buds per branch. The mean per-site ratio of the number of spruce budworm larvae to the total number of buds provided our estimation of local spruce budworm density (Sanders 1980).

Landscape-level outbreak status was calculated at multiple scales using the concentric buffer approach described above. Defoliation was summarized using polygon-based cumulative defoliation maps derived from annual aerial surveys made from 2006 to 2014 (Quebec Government 2015b). To calculate outbreak status within each buffer, we first converted the polygon defoliation maps to raster at a 30-m resolution. We next counted the total number of years between 2006 (the start of the outbreak in this area) and 2014 where each pixel was classified as either moderately or severely defoliated. Finally, we summed these values over all pixels in the buffer and standardized them by dividing by the total number of pixels in the buffer. The resulting cumulative defoliation indices for all buffers were used as predictors to model spatial variation in parasitoid communities (Table 2).

Non-environmental spatial structure

Non-environmental spatial structure in parasitoid community composition and parasitism rates models spatial autocorrelation between sites based on their geographic distance, and was included using distance-based Moran’s eigenvector maps (dbMEMs; Borcard et al. 2011). Distance-based

Table 2. Presentation of the variables considered in the study for each of the explanatory components (forest structure, spruce budworm outbreak status, and broad- and local-scale dbMEMs).

Predictor variables	Minimum	Maximum	Mean	SD
Forest structure component				
Forest land cover				
Pure or dominant balsam fir stands (%)	12.50	42.68	23.73	7.63
Pure or dominant spruce stands (%)	17.45	61.21	37.24	14.05
Pure or dominant pine stands (%)	0.05	6.74	2.03	1.55
Dominant hardwood stands (%)	0.07	11.83	5.59	3.85
Mixed forest (%)	0.22	12.81	4.18	3.50
Low vegetation (labrador tea, high grass, and/or low grass) (%)	0.39	33.65	13.96	11.46
Shannon’s diversity index based on these seven ground cover categories	1.04	1.74	1.50	0.22
Forest configuration				
Patch edge density	76.72	197.18	138.79	30.92
Mean patch area–perimeter ratio	1239.44	1387.19	1332.91	36.56
Contagion index	43.56	64.33	52.03	6.67
Spruce budworm outbreak status component				
Mean age of the spruce budworm outbreak (%)	0.29	5.91	2.62	1.55
Local spruce budworm density	0.13	0.79	0.36	0.16
Spatial component				
Broad-scale distance-based Moran’s eigenvector maps (dbMEM)	NA	NA	NA	NA
Fine-scale distance-based Moran’s eigenvector maps (dbMEM)	NA	NA	NA	NA

Notes: The range of each predictor variable in the 10,000 m buffer is given (minimum, maximum, mean, and standard deviation). NA, not applicable.

Moran's eigenvector maps are mutually orthogonal synthetic spatial variables used to model spatial variation (Legendre and Legendre 2012). Positive dbMEMs were separated into two equal groups based on eigenvalues. Distance-based Moran's eigenvector maps with large eigenvalues were used to model broad-scale spatial structure, and dbMEMs with small eigenvalues were used to model fine-scale spatial structure (Peres-Neto and Legendre 2010). Spatial variation captured using dbMEMs may represent environmental autocorrelation, dispersal limitation, or historical effects on species composition (Dray et al. 2012, Peres-Neto et al. 2012). Significant, broad-scale dbMEMs are often used as a proxy for dispersal in which the scale of spatial correlation is thought to be proportional to dispersal ability (Shurin et al. 2009). Distance-based Moran's eigenvector maps were calculated using the R package PCNM (Legendre et al. 2012).

Statistical analyses

Prior to analyzing community structure, we calculated Jaccard dissimilarity index to test for community composition changes between sampled time periods. We used the function *vegdist* within the R package VEGAN (Oksanen et al. 2015).

Variation partitioning on community data.—Variation partitioning can be used to identify the unique and shared contributions of variables (or groups of variables) to the variance in a community matrix (Legendre and Legendre 2012). Here, a matrix of per species relative parasitism rates (columns) per site (rows) served as the response variable. We considered four types of predictor variables: forest structure, outbreak status, and fine-scale and broad-scale spatial structure (dbMEMs). We used the function *varpart* within the R package VEGAN (Oksanen et al. 2015), to partition variance among these four components (Legendre and Legendre 2012). The adjusted R^2 was used to quantify variation explained (Peres-Neto et al. 2006).

Because the number of forest, spruce budworm-linked, and spatial predictors was large ($n = 10, 2$, and 10 , respectively; the first two at six different scales; total = 82), we used a two-step selection procedure to reduce the number of predictors within each component. First, for each predictor in the forest and spruce budworm-linked components, we selected from among the

six scales those that were the most informative using AIC [Akaike Information Criterion]-based stepwise selection (Burnham and Anderson 2004; Fig. 3, step 1). We combined selected forest and spruce budworm-related predictors with the fine- and broad-scale dbMEMs (spatial structure) and used the stepwise selection procedure again to identify a subset of predictors for each spatial and environmental component (Fig. 3, step 2).

Landscape-level predictors were often correlated. To reduce multi-collinearity, we computed variance inflation factors (VIFs) for each predictor within each component. When predictors were found to have a $VIF > 10$ (Borcard et al. 2011), the predictor with the highest VIF was removed. Variance inflation factors were calculated on the new set of predictors, and the evaluation procedure was repeated when necessary until all VIFs were < 10 . After filtering on the basis of VIFs, we examined correlations between retained predictors and discarded those that exhibited correlation > 0.70 , while keeping the most descriptive ones on the basis of ΔAIC (Table 3; Appendix S1). Variation partitioning analysis was then run on these selected sets of predictors (Fig. 3, step 3).

We first analyzed each of the three spruce budworm developmental stages separately to characterize and compare within-season variation in the processes driving community structure. We also analyzed all developmental stages combined to characterize the global patterns of metacommunity structure and to compare with the individual stages.

Elements of metacommunity structure.—The EMS approach examines species assemblage patterns across local communities and compares them with idealized structures (Leibold and Mikkelsen 2002) and randomness (e.g., neutral theory sensu Hubbell 2001). Membership of a metacommunity to one of these models is evaluated by analyzing three metrics: (1) coherence of species ranges, (2) species range turnover, and (3) boundary clumping (Table 1; Leibold and Mikkelsen 2002). The EMS approach proceeds as a decision tree: One first tests for coherence to distinguish non-significant (random assembly), negative (checkerboard patterns), and positive coherence (species associations; Table 1). Then, if positive coherence is identified, one tests for species turnover (Table 1). A negative turnover indicates nestedness

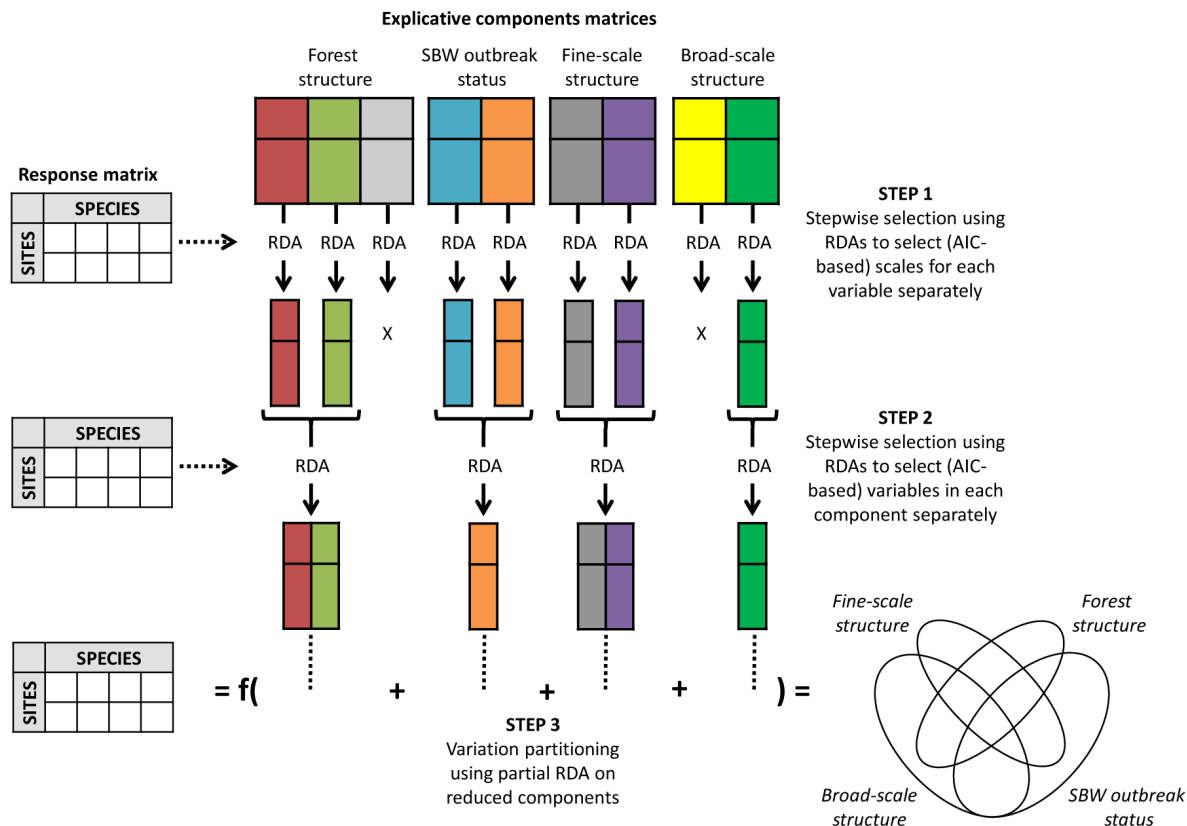


Fig. 3. Analysis pipeline for variation partitioning (VP). In the first step, and for each variable within each component (pictured here as a column in component matrices) separately, a stepwise selection based on RDA is run to select the most informative scale(s). Some variables may be discarded at this step as no scale is retained. In the second step, and for each component separately, all pre-selected variables at given scale(s) are put together to perform a new both-way stepwise procedure based on RDA. Only the most informative variables should be retained. Finally, in the third step, a variation partitioning analysis is performed on the reduced matrices of each component. The site-by-species parasitism rate matrix was used as the response variable. All this procedure was realized (1) for each of the three sampling stages separately and (2) all sampling stages combined.

(Table 1). If positive turnover is identified, one can then evaluate how environmental filtering affects species replacement along the gradient by testing for boundary clumping, which can be described as “Clementsian,” “Gleasonian,” or evenly spaced (Table 1). Elements of metacommunity structure analysis is based on a matrix of per species relative parasitism rates (columns) per site (rows), ordinated along a reciprocal averaging axis, which arranges sites based on similarities in species composition and arranges species based on similarities in distribution (i.e., correspondence analysis; Gauch 1982).

We analyzed each of our three developmental stages separately to see whether there was

within-season temporal variation in the processes driving community structure.

RESULTS

A total of 13,944 spruce budworm larvae and pupae were sampled and reared successfully from our 18 sample sites (Fig. 1). Of these samples, 22% were parasitized by a primary parasitoid. We identified 32 species of parasitoids, including 20 hymenopteran and 12 dipteran species (Appendix S2). Out of this total, 4, 12, 8, and 16 species were found at more than one site for each developmental stage (EL, LL, and P) and for all three stages combined, respectively; these species

Table 3. Spatial and landscape variables selected during the prior steps of the variation partitioning analysis for each sampling session.

Components	EL stage	LL stage	P stage	All stages
Forest structure	Spruce stands (%) [10,000 m] Patch edge density [500 m]	Balsam fir stands (%) [1000 m] Pine stands (%) [2500 m] Patch perimeter–area ratio [5000 m]	Pine stands (%) [1000 m] Low vegetation (%) [2500 m] Patch edge density [100 m] Patch perimeter–area ratio [10,000 m]	Balsam fir stands (%) [100 m] Pine stands (%) [1000 m] Low vegetation (%) [2500 m] Shannon's diversity index [10,000 m]
Spruce budworm outbreak status	Local spruce budworm density	NA	Local spruce budworm density	Local spruce budworm density
Fine-scale structure	1 dbMEM (no. 6)	NA	2 dbMEMs (nos. 4, 5)	2 dbMEMs (nos. 4, 5)
Broad-scale structure	2 dbMEMs (nos. 1, 3)	1 dbMEM (no. 1)	2 dbMEMs (nos. 1, 3)	2 dbMEMs (nos. 1, 3)

Notes: Values in square brackets represent the buffer width at which the variable was selected. For dbMEMs, values in brackets represent selected dbMEM identity, from no. 1, characterizing the broadest scale spatial variation, to no. 6, the finest scale spatial variation. Crossed out variables were discarded from final variation partitioning analysis because of co-linearity with a more explicative variable >0.70. NA, not applicable because no variable was selected; SD, standard deviation.

were included in the analysis. The average number of species per site when considering all stages combined was 10.2 (SD = 2.1). Parasitoid species abundance highly varied between sample periods, modifying community composition throughout the season (Fig. 4). Community composition varied over time and was dominated by three hymenopterans (*Glypta fumiferanae*, *Dirophanes maculicornis*, and *Apanteles fumiferanae*) and one dipteran (*Smidtia fumiferanae*), which together accounted for 88.1% of total parasitism (Fig. 4).

Variation partitioning

Final models of metacommunity structure varied between developmental stages (i.e., EL, LL, P) and when considering all stages combined (Table 3). Forest structure at multiple scales, forest diversity, and broad and fine-scale spatial structure affected parasitoid metacommunity at one or more developmental stage (Table 3). Spatial and landscape variables impacted communities at different spatial scales, from 100 to 10,000 m, which indicates the importance of applying a multi-scale

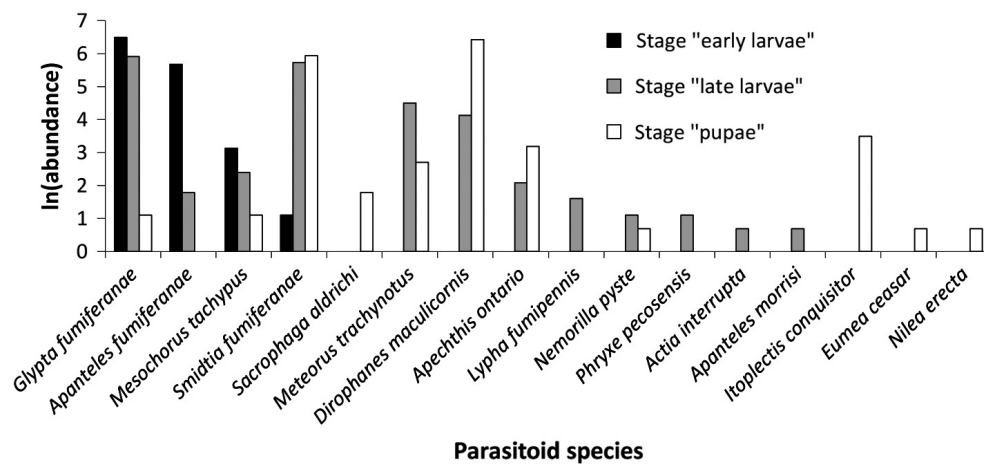


Fig. 4. Log-abundance of the 16 parasitoid species considered in our analyses over all sample locations at the three different time periods between early June and mid-July, which correspond to three different stages of spruce budworm development: early larvae (EL; black), late larvae (LL; gray), and pupae (P; white). Jaccard dissimilarity index indicated strong differences in community composition between periods: EL-LL, $J = 0.74$; EL-P, $J = 0.99$; LL-P, $J = 0.74$.

approach in environments where patches are not easily defined. We did not find any evidence for an effect of cumulative defoliation.

Total variation explained in community structure by all selected predictors was 73.4% when examining all developmental stages combined (Fig. 5a). Variation in community structure was mostly explained by the forest structure (explained variance: 60.7%). However, 64.9% of this variance was shared with the broad-scale spatial structure (Fig. 5a). The unique contributions of the fine- and broad-scale spatial structures to shape communities were relatively high (8.9% and 9.4%; Fig. 5a), while the contribution of global spruce budworm outbreak status was very low (0.3%; Fig. 5a).

Total variation explained in community structure by all selected predictors was 56.1%, 49.5%, and 60.4% in EL, LL, and P stages, respectively (Fig. 5b–d). However, the variance explained by the different components varied among stages. Communities at the EL stage were mainly structured by broad-scale spatial structure (36.0%), though 41.4% of this variance was shared with forest structure (Fig. 5b). At the LL stage,

communities were mainly structured by the forest structure (unique contribution: 39.6%; Fig. 5c). At the P stage, communities were mainly structured by the shared contribution of broad-scale spatial structure and forest structure (45.4%; Fig. 5d). Again, the unique contribution of spruce budworm outbreak status was null or very low for all stages (Fig. 5b–d).

Elements of metacommunity structure

We found evidence for significant negative coherence (more interruptions than expected by chance) in the parasitoid community at the early larvae (EL; $z = -2.65$; $P = 0.008$) and pupae (P) stages ($z = -3.75$; $P < 0.001$), and for non-significant coherence at the intermediate, LL stage ($z = -0.10$; $P = 0.918$). This suggests that most parasitoid species pairs have mutually exclusive distributions at EL and P stages (i.e., checkerboard structure; Presley et al. 2010). In contrast, the non-significant coherence at the LL stage suggests that communities are randomly structured during this time. Because we did not find any evidence for positive coherence, we did not test for turnover or boundary clumping (Leibold and Mikkelsen 2002).

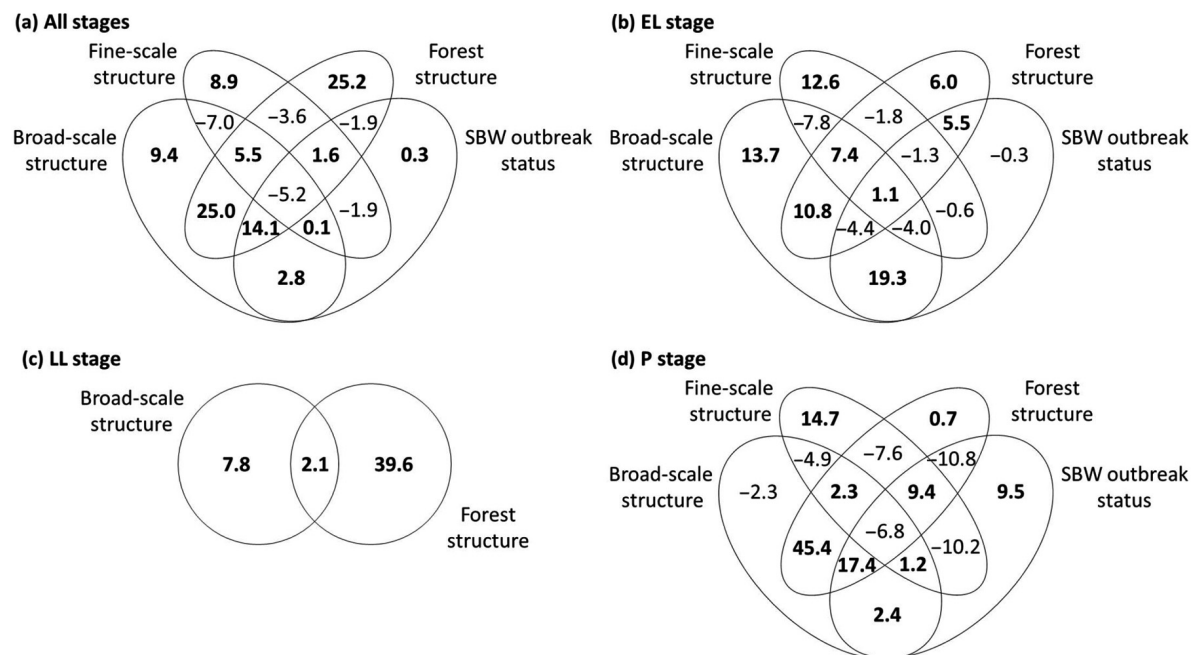


Fig. 5. Venn diagrams of the variation partitioning among selected variables in the four components: broad- and fine-scale spatial structure, forest structure, and spruce budworm outbreak status, at EL (a), LL (b), P (c), and all (d) stages. Values indicate adjusted R^2 (%).

DISCUSSION

We used two complementary methods to evaluate the relative importance of non-environmental spatial structure, forest structure, and spruce budworm outbreak status on spruce budworm-associated parasitoid metacommunity structure. We combined recent extensions of the traditional metacommunity framework to address three main challenges to studying metacommunities in forest ecosystems. First, we examined the processes shaping metacommunity structure in a continuous environment. Second, we considered the influence of environmental heterogeneity at different spatial scales in order to define, without a priori assumptions, patches in a continuous environment. And third, we examined how community structure and the processes shaping it varied through time within a single season. To our knowledge, the study of metacommunity structure in continuous environments is rare (Logue et al. 2011), and ours is one of the first to quantify the temporal variation in the relative impact of environmental and spatial factors and to integrate all three challenges at the same time.

Metacommunity structure

The theoretical paradigms that best describe parasitoid metacommunity structure varied among developmental stages. Variation partitioning analysis using all stages combined identified a mixed influence of the components identified for each separate stage (Fig. 5a).

Using the EMS approach, we determined that our parasitoid communities follow a checkerboard pattern at early (EL) and late (P) stages within a single season. Using the VP approach, we also determined that at these stages, communities are influenced by geographic distance between sample sites (broad-scale autocorrelation) and forest structure. Indeed, some variables were selected at scales up to 10,000 m, and they may be of influence at larger scales (Table 3), likely because of poor dispersal limitation at these scales. As a result, one can conclude that the patch-dynamics framework best describes metacommunity structure at these stages (Logue et al. 2011).

The checkerboard pattern identified at EL and P stages indicates mutual exclusion among species (Presley et al. 2010), which can arise due to spatial heterogeneity or competitive interactions

(Sanders et al. 2003). In the case of spruce budworm parasitoids, both processes may be present, given the patchiness of industrially exploited forest systems and known parasitoid habitat specificity (Simmons et al. 1975, Su et al. 1996, Cappuccino et al. 1998). As an example of this checkerboard pattern, we found that *Glypta fumiferanae* and *Dirophanes maculicornis* were mostly associated with homogeneous spruce forests, whereas *Apanteles fumiferanae* and *Smidtia fumiferanae* were mostly found in mixed, diversified, and fragmented forests (Appendix S3).

Competition between species can also produce checkerboard patterns. However, direct competition for host resources and space is not expected in this system as the shared host (i.e., spruce budworm larvae and pupae) is not a limiting resource (only 21.9% were parasitized) during spruce budworm outbreaks, and many parasitoid species have alternative hosts which help to avoid direct competition (O'Hara 2005, Eveleigh et al. 2007, Bennett 2008). Competition for patchily distributed habitat (e.g., floral resources) may nonetheless play a role (Amarasekare 2003). Apparent competition through indirect interactions can also affect parasitoid communities when natural enemies are shared (Holt and Lawton 1993). Indeed, parasitoids are embedded in a complex food web in which primary parasitoids may share hyperparasitoids but do not necessarily share their resources (e.g., hosts, adult food resources, habitats; Eveleigh et al. 2007). Apparent competition between species is more likely to be predator-mediated than due to direct interference (e.g., Teder et al. 2013). Eveleigh et al. (2007) showed that two of the dominant parasitoid species in the early season, *G. fumiferanae* and *A. fumiferanae*, share a high number of secondary parasitoids but not the same alternative hosts. Higher trophic levels, as well as co-evolutionary interactions between parasitoids, hosts, and their immune responses to parasitism (Béliveau et al. 2000), could represent an additional driver structuring these parasitoid metacommunities, and should represent a fascinating avenue that we will explore in the future.

In applying variation partitioning to the EL and P stages, we also identified a major combined influence of distance between sites and forest structure at large spatial scales (Fig. 5b, d). This finding indicates that at EL and P stages, geographic distance between sites is an

important determinant of community structure, and suggests that parasitoids may be dispersal limited (Meynard et al. 2013). Although dispersal dynamics can be important in shaping parasitoid communities (Couchoux et al. 2016 and references therein), their influence on observed metacommunity structures can be difficult to distinguish from other important regional processes such as latitudinal gradients.

At the LL stage, we found that community structure was influenced primarily by the local environment, at scales up to 5000 m (VP; Fig. 5c), which supports the species sorting paradigm (Logue et al. 2011). However, using the EMS analysis, we also found support for a neutral metacommunity model at this stage. Although the different results found using VP and EMS approaches may appear contradictory, we suspect that they reflect the multiple processes acting at this intermediate period and the differing sensitivity of the methods used. For example, random patterns of co-occurrence can arise when stable communities are invaded by new competitors (Sanders et al. 2003). The period of time associated with the late larval stage (i.e., mid-summer) is known to be a period of gradual change in community composition, from early spring parasitoid communities that attack young larvae to summer communities that attack mature larvae and pupae (e.g., Eveleigh et al. 2007, Eveleigh and Johns 2014). The apparently random patterns of co-occurrence we detected during this period may reflect the transitory nature of the parasitoid community at this time (Fig. 4).

In our study, the influences of broad-scale forest structure (at the LL stage, primarily) and distance between sites (at the EL and P stages, primarily) appeared to be the main components shaping parasitoid metacommunity structure. Based on these patterns, we suggest that environmental filtering and dispersal limitation are the two main ecological processes driving parasitoid metacommunity structure. This new information on how forest composition and configuration contribute to shaping parasitoid metacommunities may help to refine forest landscape management approaches to limiting spruce budworm damage. Our findings also highlight the importance of considering seasonal variation in metacommunity structure and the potentially negative consequences of ignoring this variation.

Directions for ecosystemic landscape management

Spruce budworm outbreaks have significant ecological and economic impacts (Blais 1983, Jardon et al. 2003). Improved understanding of the drivers of spruce budworm-associated parasitoid abundance and community structure has the potential to inform forest management and biocontrol strategies that aim to reduce the negative consequences of such outbreaks. Biocontrol strategies for individual parasitoid species are impractical because of the large number of species involved in the spruce budworm trophic web (Eveleigh et al. 2007) and greater benefits may be generated by managing for parasitoid habitat in a coarse-filter conservation approach (Hochberg 2000). However, to date, habitat indicators of parasitoid abundance, diversity, and efficacy are poorly known (Fraser et al. 2007). Our findings on the role of environmental filtering, on community seasonal transition primarily, and of spatial structure combined with forest structure support the idea that forest manipulations have the potential to influence parasitoid populations and maybe spruce budworm outbreak dynamics as stated by the enemies hypothesis (Cappuccino et al. 1998).

Forest composition, diversity, and configuration affected parasitoid communities in one way or another, at all sampling stages. Forest effects were present at scales up to 10,000 m (Table 3); further studies should consider the influence of spatial structure on parasitoids at least up to this scale. Greater forest diversity up this spatial scale (in terms of tree species and within-stand structures) may provide more resources for parasitoid larvae by means of alternative hosts and for adults by means of floral nectar resources (Cappuccino et al. 1999, Eveleigh et al. 2007). Although the creation of mixed balsam fir–hardwood and mixed balsam fir–spruce stands at large spatial scales has already been proposed to reduce forest losses during spruce budworm outbreaks through an increase in natural enemies (Cappuccino et al. 1998, Campbell et al. 2008, Pothier et al. 2012), the consequences for parasitoid community structure have not yet been fully explored. Our results tend to support this proposed strategy. However, explicit experimental studies should be designed in the future to directly evaluate the hypothesis.

Given the potential influence of the bird-feeder effect and of dispersal in shaping parasitoid

metacommunities, the role of habitat fragmentation to affect parasitoid movement should also be considered in forest management planning. However, parasitoid dispersal ability and its limitation by forest structure remain unknown. Landscape genetics could provide a promising avenue to respond to these objectives by identifying the environmental features that influence dispersal (i.e., affect gene flow) between populations (Manel et al. 2003).

Increased diversity and connectivity among habitat patches can increase species richness and abundance of the parasitoids and has been observed in other irruptive forest insect systems (e.g., forest tent caterpillar; Hughes et al. 2015). However, the links between forest landscape structure, community structure (diversity and abundance), dispersal, parasitism efficacy, and outbreak consequences are not straightforward. An important first step we considered in this study was to determine the mechanisms by which the environment (especially forest structure) affects the parasitoid community structure. This objective was fulfilled thanks to the integration of different challenges linked to the metacommunity framework. Next steps will require robust quantification of the functional links between spruce budworm density, parasitoid community diversity and structure, and parasitism and hyperparasitism rates, with the aim to develop predictive models of parasitoid-mediated biological control as a function of landscape context.

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SUPPORTING INFORMATION

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